INCH-POUND

MIL-C-24640/4C
26 June 1995
SUPERSEDING
MIL-C-24640/4B
31 July 1989

MILITARY SPECIFICATION SHEET

* CABLE, ELECTRIC, 600 VOLTS, LIGHT WEIGHT,
MULTI PAIR AWG 20, TYPE TTX
(INCLUDING VARIATION TYPE TTXA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-C-24640.

CONSTRUCTION
(Non-Watertight)

FIRST - 19 strand copper conductor in accordance with ASTM B-286, tinned strands in accordance with ASTM B-33. Size 20 AWG Maximum diameter 0.042" (inch).

SECOND - Component wires shall be MIL-W-81044/12 and shall be tested in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with Table VI of MIL-C-24640.

THIRD - Two component wires, one black and one white, shall be twisted together to form the required number of pairs. The pitch shall be 1" ± ¼" (inch). Identification codes applied using Method 6.

FOURTH - The required number of pairs shall be cabled together.

FIFTH - Binder tape applied helically with overlap, at manufacturer's option.

CONSTRUCTION VARIANT FOR TYPE TTX

SIXTH - Crosslinked polyolefin jacket. (See Table I for thickness).

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CONSTRUCTION VARIANT FOR TYPE TTXA

SIXTH - Crosslinked polyolefin jacket. (See Table II for thickness).

SEVENTH - Braided metal armor. Dimensions in accordance with Table II.

TABLE I. Construction details - unarmored.

PIN M24640/4	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01UN	TTX-3	3	0.038	0.296	0.320	62.1
-02UN	TTX-15	15	.044	.549	.591	221.0

TABLE II. Construction details - armored.

PIN M24640/4	Type number	Number of pairs	Jacket thickness (inch) (minimum)	Cable diameter (inch) min	Cable diameter (inch) max	Cable weight (pounds per 1000 feet) (max)
-01AN	TTXA-3	3	0.038	0.346	0.370	82.1
-02AN	TTXA-15	15	.044	.599	.641	275.4

REQUIREMENTS:

Qualification required.
Construction details: See tables I and II.

Component wire:

For the component wire the following additional tests shall be applicable in accordance with MIL-W-81044:

WIRE, RATINGS AND ADDITIONAL REQUIREMENTS:

Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300± 2°C (572± 3.6°F) for 6 hours.
Blocking: 150± 2°C (302± 3.6°F)
Color: In accordance with MIL-STD-104, class 1; white preferred
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.

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Impulse dielectric test:
Primary insulation (when tested in lieu of spark test): 6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 Khz spark test, 5.5 KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200± 2°C (392± 3.6°F) for 120 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300± 2°C (572± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150± 2°C (302± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

	Basic electrical:	Requirements:
	Conductor Resistance - ohms/1000 feet at 25°C, maximum	10.95
	Voltage withstand - volts, root mean square, minimum	
	Conductor to conductor	2000
	Conductor to shield	Not applicable
	Shield to shield	Not applicable
	Insulation resistance - megohms/1000 feet, minimum	2500
*	Conductor continuity	No failure
	Jacket flaws	Not applicable
	Group A:	
	Visual and dimensional	No failure
	Capacitance:	
	Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft), maximum	Not applicable
	Capacitance unbalance - percent, maximum	Not applicable
	Characteristic impedance - at 1 MHz, ohms	Not applicable
	Watertightness (see table XI of MIL-C-24640 for limits of water leakage)	Not applicable
	Weight	No failure
	Group B:	
	Cold bending	No failure
*	Crosslink proof test (percent maximum)	
	Jacket	50
	Drip	Not applicable
	Tear - pounds per inch thickness, minimum	35

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	Physicals (unaged)	
	Jacket	
	Tensile strength - lb/in^2, minimum	1300
	Elongation - percent, minimum	160
	Attenuation - at 3 MHz, dB/100 feet, maximum	Not applicable
	Gas flame (1 hour)	Not applicable
	Group C:	
	Physicals (aged)	
	Jacket	
	Air oven	
	Tensile strength - percent of unaged, minimum	60
	Elongation - percent of unaged, minimum	60
	Hot oil immersion	
*	Tensile strength - percent of unaged minimum	50
*	Elongation - percent of unaged, minimum	50
*	Permanence of printing (conductor) - cycles, method 6 only, minimum	25
*	Permanence of printing (jacket) - cycles, minimum	125
	Armor - (type TTXA only) - conformance to material, construction and coverage	No failure
	Shield - conformance to material, construction and coverage	Not applicable
	Heat distortion - percent of unaged, maximum	30
*	Cable sealant removability	Not applicable
	Shrinkage	No failure
	Mutual inductance - dB/1000 feet, maximum	Not applicable
	Surface transfer impedance	
	Milliohms per meter, maximum	Not applicable
	EMP response, dB, minimum	Not applicable
	Pulse response time - ms/500 feet	
	Rise time, maximum	Not applicable
	Decay time, maximum	Not applicable
*	Group D:	
	Flame propagation	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility cable	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50

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*	Smoke index, maximum	
	Jacket	25
	Fillers	45
	Insulation	10
*	Toxicity index, maximum	
	Jacket	5
	Fillers	5
	Insulation	1.5
	Durometer (jacket) - (type A) hardness, minimum	80
	Weathering (jacket)	No failure
	Fungus Resistance	No failure
	Abrasion resistance	No failure
	Gas Flame (3 hour)	Not applicable
*	Component Wire	No Failure

Unit ordering length: 500 feet nominal.

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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